

ಮಂಗಳೂರು
MANGALORE



ವಿಶ್ವವಿದ್ಯಾನಿಲಯ
UNIVERSITY

ಕ್ರಮಾಂಕ/ No. : MU/ACC/CR.35/2024-25/A8

ಕುಲಸಚಿವರ ಕಛೇರಿ

ಮಂಗಳಗಂಗೋತ್ರಿ - 574 199

Office of the Registrar
Mangalagangothri - 574 199

ದಿನಾಂಕ/Date: 03/03/2026

NOTIFICATION

Sub: Syllabus of Elective course in Statistics for IV sem B.Sc
Degree Programme -reg
Ref: Approval of the Vice-Chancellor dtd:02.03.2026

Pursuant to the above, the syllabus of Elective course in statistics for IV sem B.Sc Degree Programme prepared as per the guidelines of SEP commission is hereby notified for implementation with immediate effect subject to the ratification of academic council.

Copy of the Syllabus shall be downloaded from the Mangalore University website.

www.mangaloreuniversity.ac.in


REGISTRAR.

To:

- 1) The Registrar (Evaluation), Mangalore University, Mangalagangothri
- 2) Prof.T.PM Pakkala, Chairman, UG BOS in statistics & Retired Professor, Mangalore University
- 3) The Director, College Development council, Mangalore University.
- 4) The Principals of all affiliated/Constituent Colleges of Mangalore University .
- 5) The Assistant Registrar/The Superintendent, Academic Section, O/o the Registrar, Mangalore University.
- 6) The Nodal Officer, Computer Centre, Mangalore University – with a request to publish in the Website.
- 7) Guard File.

Discipline Elective course in Statistics for IV sem B Sc Programme

Course Title: **Reliability**

Course Objective

This course aims to introduce students to the fundamental concepts of reliability theory and life distributions, focusing on reliability functions, hazard rates, and their interrelationships. It develops understanding of component and system reliability, ageing properties (IFR, IFRA, NBU, DMRL, NBUE), and important measures such as mean time to failure and median life. The course equips students with the theoretical foundation required to analyze reliability models and apply them to practical problems in industry and applied sciences.

Course Outcomes

After successful completion of the course, students will be able to:

- 1:** Define and explain key concepts in reliability theory, including reliability function, failure distribution, density function, and hazard rate.
- 2:** Establish and interpret the mathematical relationships between reliability function, failure rate, and related life distribution functions.
- 3:** Analyze and compute reliability measures for standard life distributions such as Exponential, Weibull, and Gamma distributions.
- 4:** Evaluate the reliability of component and coherent systems and determine system reliability using appropriate models.
- 5:** Classify life distributions based on ageing properties (IFR, IFRA, NBU, DMRL, NBUE), interpret the Bathtub curve, and compute measures such as mean time to failure and median life.

Course code	Course title	Instruction hours/week	Duration of the exam	Max Marks			Credits
				IA	Exam	Total	
BSCSTES401	Reliability	2	2	10	40	50	2

Reliability: Total 24 hours

Unit 1

Introduction, definition of Reliability, Life distributions. Basic functions in Reliability reliability function, failure distribution function, failure density function, hazard rate relationship between functions. Examples from exponential, Weibull, gamma.

Unit 2

Components and systems of Reliability system, coherent systems, reliability of coherent systems, Life distributions - reliability function, hazard rate, common life distributions -

Notions of ageing – IFR, IFRA, NBU, DMRL and NBUE classes. Mean time to failure and median of the random variable. Bathtub Curve.

References:

- [1] N. Ravi Chandran *Reliability Theory*, Wiley Eastern
- [2] B D Sivazlian and L E Stanfel (1975) *Analysis of systems of Operations Research*, Prentice-Hall Inc.
- [3] Marvin Rausand and ArnltotHeyland (2004) *System Reliability Theory*, Wiley Interscience
- [4] R. E. Barlow and F Proschan (1975) *Statistical theory of reliability and life testing*, HoltRinhert, Winston